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Partial Derivatives of the Solution to the Lambert Boundary Value Problem

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Two methods for deriving first-order partial derivatives of the outputs with respect to the inputs of the Lambert boundary value problem are presented. The first method assumes the Lambert problem is solved via the universal vercosine formulation. Taking advantage of inherent symmetries and intermediate variables, the derivatives are expressed in a computationally efficient form. The typical added cost of computing these partials is found to be ∼15 to 35% of the Lambert computed cost. A second set of the same partial derivatives is derived from the fundamental perturbation matrix, also known as the state transition matrix of the Keplerian initial value problem. The equations are formulated in terms of Battin's partitions of the state transition matrix and its adjoint. This alternative approach works with any Lambert formulation, including one that solves a perturbed Lambert problem, subject to the availability of the associated state transition matrix. The analytic partial derivatives enable fast trajectory optimization formulations that implicitly enforce continuity constraints via embedded Lambert problems.

Nomenclature

$\boldsymbol{a}, A$	=	notation for vector (column) and matrix, respectively
a_b	=	$\partial a/\partial b$, where a and b are general variables
d	=	transfer angle parameter ($+1$ for $0 < \theta < \pi$, -1 for $\pi < \theta < 2\pi$)
E	=	eccentric anomaly
F	=	hyperbolic eccentric anomaly
f	=	Lagrange function (part of f and g functions)
g	=	Lagrange function (part of f and g functions), TU
k	=	universal variable for solving Lambert equation using vercosine formulation
k^*	=	value of k corresponding to T^*, TU
LU	=	length unit
N_{rev}	=	number of revolutions
q	=	generic variable name representing a scalar function
$[\boldsymbol{r}_1, \boldsymbol{r}_2]$	=	initial and final position vectors, respectively (input to Lambert problem), LU
S	=	geometry parameter appearing in Lambert equation, TU
T_p	=	parabolic time of flight, TU
T^*	=	target time of flight (input to Lambert problem), TU
$[t_1, t_2]$	=	initial and final times, respectively, TU
TU	=	time unit
$[\boldsymbol{v}_1, \boldsymbol{v}_2]$	=	initial and final velocity vectors, respectively (output to Lambert problem), LU/TU
W	=	auxiliary function appearing in Lambert equation
$[\delta, d]$	=	variation and differential operators, respectively
ϵ	=	small numerical value (e.g., 0.02)
θ	=	transfer angle
μ	=	gravitational parameter of the primary, $(\text{LU})^3/(\text{TU})^2$
τ	=	geometry parameter appearing in Lambert equation
$\Phi(t_2, t_1)$	=	state transition matrix from time t_1 to t_2, with first quadrant units of TU and third quadrant units of $(\text{TU})^{-1}$

Subscript

LT	=	expression subjected to the Lambert targeting constraints

I. Introduction

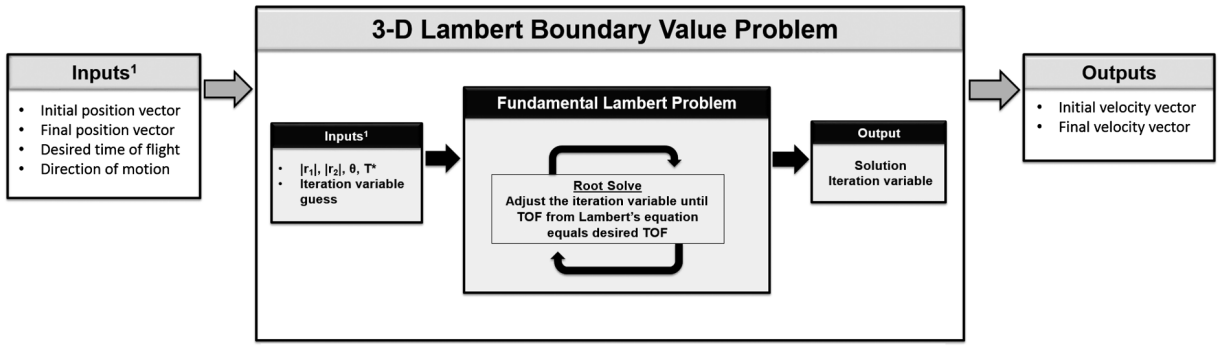
LAMBERT'S problem is one of the most extensively studied problems in space-flight mechanics and is the subject of a large volume of research spanning over several decades [1–7]. The solution to the Lambert problem acts as a building-block algorithm for various problems like trajectory design and maneuver optimization [8–15], global optimization [16–18], tour design [9,19–24], and orbit determination [25,26].

Although the partial derivatives of the Kepler initial value problem solution play a critical and well-understood role in a variety of astrodynamics algorithms, the partial derivatives of the Lambert boundary value problem solution have received far less attention. A variety of preliminary mission design search methods requires the solution to large numbers of Lambert problems [9,11,12,14,20,21,24]. However, when transitioning to the direct optimization of a single trajectory, most formulations use initial value propagation schemes (Kepler or numerical integration) and some form of shooting algorithm to explicitly enforce boundary value constraints [27–30]. If such a formulation is to be solved using gradient-based methods, the solver will require partial derivatives of solutions to the Kepler initial value problem (not the Lambert boundary value problem). A less common,

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[1]Note that the number of revolutions must also be specified, and if multiple revolutions are desired, the user must also specify a short period or a long period.

Fig. 1 Diagram of the Lambert problem and associated inputs and outputs.

alternative optimization formulation [31] embeds the Lambert problem solution inside each major iteration to implicitly enforce the boundary value constraints without the need for a shooting method. If such a formulation is to be solved using gradient-based methods, the solver will require partial derivatives of solutions to the Lambert boundary value problem. See Fig. 1, where the partials of interest are the sensitivity of the output (far right box) with respect to the input (far left box).

Note that Lambert's original theorem states that the time of flight is functionally dependent only on the semimajor axis, the sum of the magnitudes r_1 and r_2, and the chord length connecting $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ ([4] p. 276). In the current study, a universal variable k replaces the semimajor axis and is used for the iteration variable, such that the Lambert equation for time of flight equals a desired time of flight T^*. The "Lambert problem" (not to be confused with Lambert's theorem) is defined as the boundary value problem, with the inputs and outputs given in Fig. 1. The fundamental scalar problem deals with the one-dimensional (1-D) root solve, whereas the three-dimensional (3-D) problem deals with the more general problem with vector inputs and outputs.

The inner loop of the diagram in Fig. 1 is an iterative procedure involving a 1-D root solve. Therefore, computing the sensitivities of the outputs to the inputs is not as straightforward as computing the partials of an explicit function. To avoid this complication, the desired partials are commonly computed via central differencing due to its ease of implementation. Such an approach is practical but suffers from slow runtimes and low accuracies. However, most local optimization algorithms [15–18,32] benefit significantly from accurate partial derivatives, especially in the more challenging and highly nonlinear spacecraft trajectory problems. Complex step differentiation and automatic differentiation resolve the accuracy problems but retain the speed penalties and require significant implementation effort.

In the current study, the partials of interest are derived in detail in two ways. In the first solution method, the expressions are derived using the recently proposed vercosine (hereafter referred to as vercosine) Lambert formulation [1]. This new Lambert solver takes advantage of geometry-based parameters (S, τ) to simplify the universal formulation of the Lambert equation. This equation, defined as a function of the universal variable k, is shown to have simplified derivative expressions and requires only a single transcendental function W evaluation. Apart from enjoying runtime improvements over the current state-of-the-art Gooding's method [1], the new vercosine formulation also leads to an efficient representation of the partial derivatives. The extension of the vercosine formulation to include the solution partial derivatives is a main contribution of the current study. The process involves taking derivatives of the classic f and g functions (also called Lagrange coefficients; see Bate et al. [33] pp. 218–219) and applying the chain rule, taking into account various implicit dependencies. The decoupling of the geometric (S, τ) and iterative terms (k) in the Lambert formulation simplifies the derivatives of the universal variable k with respect to the fundamental scalar dependencies (r_1, r_2, θ, T^*).

In addition to this direct approach, a second, more general method of computing the partials is derived using a constrained linearization and the classic state transition matrix (STM) of the Kepler initial value problem. The results serve as validation for the first approach and represent a standalone contribution. A similar method for a custom problem has been previously proposed [31] and was used to optimize Galileo's trajectory [13]. It is noted that D'Amario et al. [31] gave explicit formulas, without derivation, for the partials of the solution to the perturbed Lambert problem, provided a state transition matrix was available. In the current study, new equivalent formulas are carefully derived in terms of the partitions of the fundamental perturbation matrices, first introduced by Battin, as stated in his seminal book ([4] p. 467). Once the Lambert problem is solved, this second approach requires a user to obtain the STM of the Kepler problem by any means available, such as those found in work by Battin ([4] p. 467) or Pitkin [34]. Note that the Kepler STM calculation is not iterative like the Kepler problem itself. In the current study, Battin's formulation for computing the STM is used. This second method that uses partitions of the STM is valid for perturbed Lambert problems (i.e., boundary value problems with non-Keplerian dynamics).

It is emphasized that both solution methods give accurate formulas for the first-order partials that can be used for linearized analyses. The quality of the partials should not be confused with the validity of a linearized model. Any optimization or targeting method that performs a local linearization is of course limited to sufficiently small update increments such that the linearized model is a good approximation.

In this study, all first-order partial derivatives of the Lambert problem solutions are derived and documented using both methods. A large-scale simulation of tens of millions of Lambert calls is performed to quantify the cost and accuracy of computing these partial derivatives. The next section gives a brief summary of the Lambert formulation necessary for derivation of the partials using the first solution technique.

II. Multiple-Revolution Lambert Formulation

This section summarizes the Lambert formulation based on the new universal variable [1] k defined as follows:

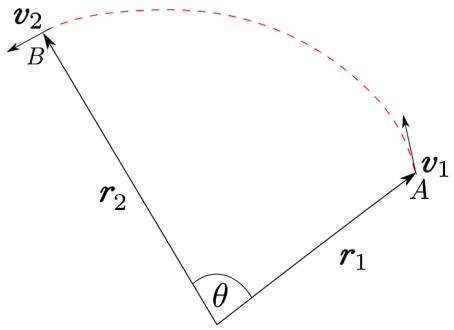
$$k^2 = \text{vercosine}(\Delta E) = 1 + \cos(\Delta E) \tag{1}$$

where $\Delta E = E_2 - E_1$ represents the change in eccentric anomaly. Note that the hyperbolic anomaly F and the eccentric anomaly E are related via the definition $F = \sqrt{-1}E = iE$.

Figure 2 depicts a general diagram of the Lambert problem. Vectors $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ are the bounding position vectors; T^* stands for the time of flight for the transfer; and θ is the transfer angle between the two position vectors. All possible arcs connecting the two points A and B with the required T^*, and which satisfy Keplerian motion, are possible Lambert solutions. There are $2N^D + 1$ direct solutions and $2N^R + 1$ retrograde solutions, where N^D and N^R are the maximum number of revolutions possible for direct and retrograde transfers, respectively, given a set of input parameters.

As Keplerian motion is confined in a plane, vectors $\boldsymbol{r}_2$ and $\boldsymbol{v}_2$ can be expressed as a linear combination of $\boldsymbol{r}_1$ and $\boldsymbol{v}_1$:

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v_1 and v_2 are obtained after solving the Lambert problem

Fig. 2 General problem geometry.

$$\boldsymbol{r}_2 = f\boldsymbol{r}_1 + g\boldsymbol{v}_1 \tag{2}$$

$$\boldsymbol{v}_2 = \dot{f}\boldsymbol{r}_1 + \dot{g}\boldsymbol{v}_1 \tag{3}$$

where f and g are the Lagrange "f and g functions" [33], respectively. The vercosine formulation defines two geometry-based parameters: S and τ. The parameter τ varies between

$$\left[\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$$

and is a nondimensional function of r_1, r_2, and θ:

$$\tau = d\frac{\sqrt{r_1 r_2[1+\cos(\theta)]}}{r_1 + r_2}, \qquad d = \begin{cases} +1 & 0 \le \theta \le \pi \\ -1 & \pi < \theta < 2\pi \end{cases} \tag{4}$$

The parameter S has a unit of time and is defined as

$$S = \sqrt{\frac{(r_1 + r_2)^3}{\mu}} \tag{5}$$

Next, the Lambert time-of-flight (TOF) equation is found to be a function of S, τ, and the universal variable k:

$$\mathrm{TOF}(k, S, \tau) = S\sqrt{1-k\tau}[\tau + (1-k\tau)W] \tag{6}$$

where W is given by the following expression:

$$W(k) = \begin{cases} \dfrac{(1-\mathrm{sgn}(k))\pi + \mathrm{sgn}(k)\cos^{-1}(1-m) + 2\pi N}{\sqrt{m^3}} - \dfrac{k}{m} & -\sqrt{2} < k < \sqrt{2} - \epsilon \text{ (elliptical orbits)} \\ \dfrac{-\cosh^{-1}(1-m)}{\sqrt{-m^3}} - \dfrac{k}{m} & k > \sqrt{2} + \epsilon \text{ (hyperbolic orbits)} \\ W_s & \sqrt{2} - \epsilon \le k \le \sqrt{2} + \epsilon \ (N = 0) \end{cases} \tag{7}$$

where $m = 2 - k^2$, and W_s is given by the following series:

$$\begin{aligned} W_s = {} & \frac{\sqrt{2}}{3} - \frac{v}{5} + \frac{2}{35}\sqrt{2}v^2 - \frac{2}{63}v^3 + \frac{2}{231}\sqrt{2}v^4 - \frac{2}{429}v^5 + \frac{8}{6435}\sqrt{2}v^6 \\ & - \frac{8}{12155}v^7 + \frac{8}{46,189}\sqrt{2}v^8 - \frac{8}{88,179}v^9 + \frac{16}{6,76,039}\sqrt{2}v^{10} \\ & - \frac{16}{1,300,075}v^{11} + \frac{16}{5,014,575}\sqrt{2}v^{12} - \frac{16}{9,694,845}v^{13} \\ & + \frac{128}{300,540,195}\sqrt{2}v^{14} - \frac{128}{583,401,555}v^{15} + \frac{128}{2,268,783,825}\sqrt{2}v^{16} \\ & + O(v^{33/2}) \\ v = {} & k - \sqrt{2} \end{aligned} \tag{8}$$

Note that the number of revolutions is accounted for in the definition of W. Equation (6) is the Lambert TOF equation that is root solved in the inner loop of Fig. 1. For a given value of k, the f and g functions are used to calculate the initial and final velocity vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ as follows:

$$f = 1 - \frac{(1-k\tau)(r_1 + r_2)}{r_1} \tag{9}$$

$$g = S\tau\sqrt{(1-k\tau)} \tag{10}$$

$$\dot{g} = 1 - \frac{(1-k\tau)(r_1 + r_2)}{r_2} \tag{11}$$

$$\boldsymbol{v}_1 = \frac{\boldsymbol{r}_2 - f\boldsymbol{r}_1}{g} \tag{12}$$

$$\boldsymbol{v}_2 = \frac{\dot{g}\boldsymbol{r}_2 - \boldsymbol{r}_1}{g} \tag{13}$$

Note that Eqs. (9–11) are corrections of a typographical error, as given in [1].

III. Lambert Solution Partials Using the Direct Approach

The partials of the Lambert solution are defined as partials of $\boldsymbol{y}$ with respect to $\boldsymbol{p}$, where $\boldsymbol{y}$ and $\boldsymbol{p}$ are given as follows (and illustrated in Fig. 1):

$$\boldsymbol{y} = \begin{pmatrix} \boldsymbol{v}_1 \\ \boldsymbol{v}_2 \end{pmatrix} \tag{14}$$

$$\boldsymbol{p} = \begin{pmatrix} \boldsymbol{r}_1 \\ \boldsymbol{r}_2 \\ T^* \end{pmatrix} \tag{15}$$

To obtain $\mathrm{d}\boldsymbol{y}/\partial\boldsymbol{p}$ we need to compute $\mathrm{d}q/\mathrm{d}\boldsymbol{p}$, where q can be either f, g, or $\dot{g}$ [see Eqs. (12) and (13)]:

$$q \in \{f, g, \dot{g}\} \tag{16}$$

The scalar q itself is a function of a vector $\boldsymbol{p}$ and the Lambert iteration variable k. Using the chain rule, $\mathrm{d}q/\mathrm{d}\boldsymbol{p}$ can be computed as follows:

$$\frac{\mathrm{d}q}{\mathrm{d}\boldsymbol{p}} = q_p + q_k k_p \tag{17}$$

where k is the root-solve parameter of the Lambert equation, and k_p is given as

$$k_p = k_x \boldsymbol{x}_p \tag{18}$$

$$\boldsymbol{x} = [r_1, r_2, \theta, T^*]^T \tag{19}$$

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Note that the scalar parameters of the root-solve equation [Eq. (20)] are grouped into the single vector $\boldsymbol{x}$. The partials using the variable k are derived in the next section.

A. Partials of the Iteration Variable with Respect to Basic Inputs: k_x

The partials given by k_x represent the fundamental partials of the Lambert problem solution and answer the fundamental question "What is the sensitivity of the the parameter k after a successful root solve with respect to the inputs of the problem?" Given $\boldsymbol{x}$, the solution to the Lambert problem is found via a one-dimensional root solve using the iteration variable k of the function L, given as follows:

$$L(k) = \text{TOF}[k, S(\boldsymbol{x}), \tau(\boldsymbol{x})] - T^* \tag{20}$$

at the solution

$$L(k^*) = 0 = \text{TOF}[k^*, S(\boldsymbol{x}), \tau(\boldsymbol{x})] - T^* \tag{21}$$

where k^* is the root-solved value of k. Next, taking the derivative of the preceding equation with respect to the vector $\boldsymbol{x}$, we get the following:

$$\frac{\mathrm{d}L}{\mathrm{d}\boldsymbol{x}} = \frac{\mathrm{dTOF}}{\mathrm{d}\boldsymbol{x}} - \frac{\mathrm{d}T^*}{\mathrm{d}\boldsymbol{x}} \tag{22}$$

where

$$\frac{\mathrm{dTOF}}{\mathrm{d}\boldsymbol{x}} = \text{TOF}_k k_{\boldsymbol{x}} + \text{TOF}_S S_{\boldsymbol{x}} + \text{TOF}_\tau \tau_{\boldsymbol{x}} \tag{23}$$

and

$$\text{TOF}_S = \frac{T^*}{S}$$

$$\text{TOF}_\tau = \frac{T^*}{\tau\gamma}, \qquad \gamma = \frac{1+cW}{1-\frac{k^*}{2c}[3cW+1]}, \qquad c = \frac{1-k^*\tau}{\tau}$$

$$\text{TOF}_k = \frac{-T^*}{2c} + S\tau\sqrt{c\tau}(W_k c - W)$$

W_k is the partial of W from Eqs. (7) and (8) with respect to k:

$$W_k = \begin{cases} \frac{-2+3Wk}{m} & k < \sqrt{2}-\epsilon, k > \sqrt{2}+\epsilon \\ \frac{\partial W_\pi}{\partial k} & \sqrt{2}-\epsilon < k < \sqrt{2}+\epsilon \end{cases} \tag{24}$$

Note that TOF_k and W_k are computed and saved during the root solve of L. For all subsequent sections, we drop the superscript $*$ over k for presentation purposes. Substituting Eq. (23) into Eq. (22), and equating the result to zero, leads to

$$k_{\boldsymbol{x}} = \frac{-T^*}{\text{TOF}_k}\left[-\frac{\mathrm{d}T^*}{\mathrm{d}\boldsymbol{x}}\frac{1}{T^*} + \frac{S_{\boldsymbol{x}}}{S} + \frac{\tau_{\boldsymbol{x}}}{\tau\gamma}\right] \tag{25}$$

The expressions for the partials $T^*_{\boldsymbol{x}}$, $S_{\boldsymbol{x}}$, and $\tau_{\boldsymbol{x}}$ are given as follows:

$$\frac{\mathrm{d}T^*}{\mathrm{d}\boldsymbol{x}} = [0,0,0,1]$$

$$S_{\boldsymbol{x}} = [S_{r_1}, S_{r_2}, 0, 0]$$

$$\tau_{\boldsymbol{x}} = [\tau_{r_1}, \tau_{r_2}, \tau_\theta, 0]$$

$$S_{r_1} = S_{r_2} = \frac{3S}{2(r_1+r_2)}$$

$$\tau_{r_1} = \tau\left[\frac{1}{2r_1} - \frac{1}{r_1+r_2}\right], \qquad \tau_{r_2} = \tau\left[\frac{1}{2r_2} - \frac{1}{r_1+r_2}\right]$$

$$\tau_\theta = \frac{-\tau\sin(\theta)}{2(1+\cos(\theta))} \tag{26}$$

B. Self-Partials: x_p, S_p, and τ_p

The self-partials $\boldsymbol{x_p}$ map the scalar partial derivatives of k to their corresponding vector versions. They are simply defined as follows:

$$\boldsymbol{x_p} = [r_{1p}^T, r_{2p}^T, \theta_p^T, T_p^{*T}]^T \tag{27}$$

where $\boldsymbol{x_p}$ is a 4×7 matrix, and its components are calculated as follows:

$$r_{1p} = [\hat{\boldsymbol{r}}_1^T, 0, 0, 0, 0]$$

$$r_{2p} = [0, 0, 0, \hat{\boldsymbol{r}}_2^T, 0]$$

$$\theta_p = \left[\frac{\hat{\boldsymbol{r}}_1^T\cos(\theta) - \hat{\boldsymbol{r}}_2^T}{r_1\sin(\theta)}, \frac{\hat{\boldsymbol{r}}_2^T\cos(\theta) - \hat{\boldsymbol{r}}_1^T}{r_2\sin(\theta)}, 0\right]$$

$$T_p^* = [0,0,0,0,0,0,1] \tag{28}$$

where $\hat{\boldsymbol{r}}_1$ and $\hat{\boldsymbol{r}}_2$ are unit vectors corresponding to $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, respectively. The self-partials, S_p and τ_p, are given as

$$S_p = S_{\boldsymbol{x}}\boldsymbol{x_p} \quad \tau_p = \tau_{\boldsymbol{x}}\boldsymbol{x_p} \tag{29}$$

where the partials $S_{\boldsymbol{x}}$ and $\tau_{\boldsymbol{x}}$ are given in Eq. (26). The next section summarizes the partials required to compute the chain rule.

C. Chain Rule Partials: q_p and q_k

The partials to complete the chain rule in Eq. (17) are given as follows:

$$q_p \in \{f_p, g_p, \dot{g}_p\} \quad q_k \in \{f_k, g_k, \dot{g}_k\} \tag{30}$$

Obtaining the partials in Eq. (30) requires the partials of f, g, and $\dot{g}$ with respect to $\boldsymbol{p}$:

$$f_p = (f_{r_1}r_{1p} + f_{r_2}r_{2p} + f_\tau\tau_p)$$

$$g_p = (g_S S_p + g_\tau\tau_p)$$

$$\dot{g}_p = (\dot{g}_{r_1}r_{1p} + \dot{g}_{r_2}r_{2p} + \dot{g}_\tau\tau_p)$$

$$f_{r_1} = (1-f)\left[\frac{1}{r_1} - \frac{1}{r_1+r_2}\right], \quad f_{r_2} = \frac{-(1-f)}{r_1+r_2}, \quad \dot{g}_{r_1} = \frac{-(1-\dot{g})}{(r_1+r_2)},$$

$$\dot{g}_{r_2} = (1-\dot{g})\left[\frac{1}{r_2} - \frac{1}{r_1+r_2}\right]$$

$$f_\tau = \frac{k(r_1+r_2)}{r_1}, \quad \dot{g}_\tau = \frac{k(r_1+r_2)}{r_2}$$

$$g_\tau = g\left[\frac{1}{\tau} - \frac{k}{2c\tau}\right], \quad g_S = \frac{g}{S} \tag{31}$$

where c has been computed before as an intermediate variable in Eq. (23). Computing q_k requires the partials of the f and g functions with respect to k:

$$f_k = \frac{\tau(r_1+r_2)}{r_1}, \qquad \dot{g}_k = \frac{\tau(r_1+r_2)}{r_2}, \qquad g_k = -\frac{g}{2c} \tag{32}$$

Finally, the derivatives defined in Eq. (17) can be computed. The next section addresses the partials of the velocity vectors.

D. Partials of the Velocity Vectors

Finally, the partial of the velocities $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ with respect to the vector $\boldsymbol{p}$ can be computed by differentiating Eqs. (12) and (13) along with the Eq. (17):

$$\left[\frac{\partial \boldsymbol{v}_1}{\partial \boldsymbol{p}}\right]_{\text{LT}} = \frac{\boldsymbol{r}_2 - [(\mathrm{d}f/\mathrm{d}\boldsymbol{p})\boldsymbol{r}_1 + f\boldsymbol{r}_{1p}] - \boldsymbol{v}_1 g_p}{g} \tag{33}$$

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$$\left[\frac{\partial \boldsymbol{v}_2}{\partial \boldsymbol{p}}\right]_{\mathrm{LT}} = -\frac{\boldsymbol{r}_1 - [(\mathrm{d}\dot{g}/\mathrm{d}p)\boldsymbol{r}_2 + \dot{g}\boldsymbol{r}_{2p}] - \boldsymbol{v}_2 g_p}{g} \tag{34}$$

where the LT subscript indicates the partial is subject to the Lambert targeting constraint. For completeness, the partials of the velocities $(\boldsymbol{v}_1, \boldsymbol{v}_2)$ with respect to the times t_1 and t_2 are symmetric and given as

$$\left[\frac{\partial \boldsymbol{v}_1}{\partial t_2}\right]_{\mathrm{LT}} = \left[\frac{\partial \boldsymbol{v}_1}{\partial T^*}\right]_{\mathrm{LT}}, \qquad \left[\frac{\partial \boldsymbol{v}_1}{\partial t_1}\right]_{\mathrm{LT}} = -\left[\frac{\partial \boldsymbol{v}_1}{\partial T^*}\right]_{\mathrm{LT}} \tag{35}$$

$$\left[\frac{\partial \boldsymbol{v}_2}{\partial t_2}\right]_{\mathrm{LT}} = \left[\frac{\partial \boldsymbol{v}_2}{\partial T^*}\right]_{\mathrm{LT}}, \qquad \left[\frac{\partial \boldsymbol{v}_2}{\partial t_1}\right]_{\mathrm{LT}} = -\left[\frac{\partial \boldsymbol{v}_2}{\partial T^*}\right]_{\mathrm{LT}} \tag{36}$$

E. Symmetry in Partials

Careful investigation of Eqs. (33) and (34) reveals that there exist three kinds of symmetry. First, there exists cross symmetry between the partial matrices $\partial \boldsymbol{v}_1/\partial \boldsymbol{r}_2$ and $\partial \boldsymbol{v}_2/\partial \boldsymbol{r}_1$ [defined by Eq. (37)] such that only one of them needs to computed:

$$\frac{\partial \boldsymbol{v}_1}{\partial \boldsymbol{r}_2} = -\frac{\partial \boldsymbol{v}_2{}^T}{\partial \boldsymbol{r}_1} \tag{37}$$

Second, the partial matrices $\partial \boldsymbol{v}_1/\partial \boldsymbol{r}_1$ and $\partial \boldsymbol{v}_2/\partial \boldsymbol{r}_2$ are both symmetric. Third, the partials of each velocity with respect to the times t_1 and t_2 are identical with a simple sign switch [see Eqs. (35) and (36)]. By taking advantage of these symmetries, the number of partials required to be computed is reduced from 48 to 27. Further validation of the symmetries is apparent in the resulting expressions from the second solution method, as discussed next.

IV. Lambert Solution Partials Using the Kepler STM

A second method is derived for the analytic position and time-of-flight partials of the Lambert problem using the Kepler STM and linearization of initial and final states. For clarity, we define the Kepler STM as synonymous with Battin's fundamental perturbation matrices and not the f, g, $\dot{f}$, and $\dot{g}$ expressions (Battin [4] p. 130). This second method to compute the Lambert solution partials is general because it is uncoupled from a particular Lambert algorithm and a Kepler STM algorithm. In fact, the method even applies to perturbed dynamics, as long as the same dynamics are embedded in the solved boundary value problem and associated STM. In this study, the Kepler STM is computed using Battin's expressions of the universal functions $U_1 - U_5$, which are provided next. Note that this second method does not use a root solve of the Lambert equation. Therefore, T^* and TOF have identical meanings in the context of the Kepler STM approach.

A. Kepler State Transition Matrix in Terms of the Universal Functions

The Lagrange functions can be stated in terms of the universal functions $U_1 - U_5$. Battin's formulation for f, g and $\dot{f}$ on p. 464 of his text [4] can be inverted to solve for U_1, U_2, and U_3; equation 4.108 ([4] p. 189) provides U_4; and the identity given in equation 4.76 ([4] p. 176) can be solved for U_5:

$$U_1 = -\frac{r_2 r_1 \dot{f}}{\sqrt{\mu}} \tag{38}$$

$$U_2 = r_1(1-f) \tag{39}$$

$$U_3 = \sqrt{\mu}(\mathrm{TOF} - g) \tag{40}$$

$$U_4 = U_1 U_3 - \frac{1}{2}(U_2^2 - \alpha U_3^2) \tag{41}$$

$$U_5 = \frac{\chi^3/6 - U_3}{\alpha} \tag{42}$$

where

$$\sigma_1 = \boldsymbol{r}_1^T \boldsymbol{v}_1/\sqrt{\mu}, \qquad \sigma_2 = \boldsymbol{r}_2^T \boldsymbol{v}_2/\sqrt{\mu} \tag{43}$$

$$\alpha = 2/r_1 - v_1^2/\mu, \qquad \chi = \alpha\sqrt{\mu}\mathrm{TOF} + \sigma_2 - \sigma_1 \tag{44}$$

Note that $\alpha = \frac{1}{a}$, where a is the semimajor axis. Assuming a Lambert algorithm is implemented, the universal functions $U_1 - U_5$ are known via Eqs. (38–42) because f, g, $\dot{f}$, and $\dot{g}$ are known.

Special care must be taken in determining U_5 because of the singularity when α is close to zero. The singularity occurs when the trajectory is near parabolic. For the near-parabolic case, a continued fraction for U_5 (Battin [4] p. 190) is implemented instead. An efficient implementation of the continued fraction is given as the G variable by Shepperd [35]. For extremely high flight time cases, the Battin ([4] p. 190) and Shepperd [25] implementations suffer from an unresolved precision problem where the continued fraction form of U_5 becomes very large and unreliable. Because those cases have large but finite semimajor axes, the U_5 form given in Eq. (42) suffices.

Battin ([4] p. 467) states the Kepler STM in terms of the universal functions U_1–U_5:

$$\Phi(t_2, t_1) = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{bmatrix} \tag{45}$$

$$\begin{aligned} \boldsymbol{A} &= \frac{\partial \boldsymbol{r}_2}{\partial \boldsymbol{r}_1} \\ &= \frac{r_2}{\mu}(\boldsymbol{v}_2 - \boldsymbol{v}_1)(\boldsymbol{v}_2 - \boldsymbol{v}_1)^T + \frac{1}{r_1^3}[r_1(1-f)\boldsymbol{r}_2\boldsymbol{r}_1^T + \bar{C}\boldsymbol{v}_2\boldsymbol{r}_1^T] + f\boldsymbol{I} \end{aligned} \tag{46}$$

$$\boldsymbol{B} = \frac{\partial \boldsymbol{r}_2}{\partial \boldsymbol{v}_1} = \frac{r_1}{\mu}(1-f)[(\boldsymbol{r}_2 - \boldsymbol{r}_1)\boldsymbol{v}_1^T - (\boldsymbol{v}_2 - \boldsymbol{v}_1)\boldsymbol{r}_1^T] + \frac{\bar{C}}{\mu}\boldsymbol{v}_2\boldsymbol{v}_1^T + g\boldsymbol{I} \tag{47}$$

$$\begin{aligned} \boldsymbol{C} = \frac{\partial \boldsymbol{v}_2}{\partial \boldsymbol{r}_1} &= -\frac{1}{r_1^2}(\boldsymbol{v}_2 - \boldsymbol{v}_1)\boldsymbol{r}_1^T - \frac{1}{r_2^2}\boldsymbol{r}_2(\boldsymbol{v}_2 - \boldsymbol{v}_1)^T \\ &+ \dot{f}\left[\boldsymbol{I} - \frac{1}{r_2^2}\boldsymbol{r}_2\boldsymbol{r}_2^T + \frac{1}{\mu r_2}(\boldsymbol{r}_2\boldsymbol{v}_2^T - \boldsymbol{v}_2\boldsymbol{r}_2^T)\boldsymbol{r}_2(\boldsymbol{v}_2 - \boldsymbol{v}_1)^T\right] - \frac{\mu\bar{C}}{r_2^3 r_1^3}\boldsymbol{r}_2\boldsymbol{r}_1^T \end{aligned} \tag{48}$$

$$\boldsymbol{D} = \frac{\partial \boldsymbol{v}_2}{\partial \boldsymbol{v}_1} = \frac{r_1}{\mu}(\boldsymbol{v}_2 - \boldsymbol{v}_1)(\boldsymbol{v}_2 - \boldsymbol{v}_1)^T + \frac{1}{r_2^3}[r_1(1-f)\boldsymbol{r}_2\boldsymbol{r}_1^T - \bar{C}\boldsymbol{r}_2\boldsymbol{v}_1^T] + \dot{g}\boldsymbol{I} \tag{49}$$

where $\bar{C}$ is defined as

$$\bar{C} = \frac{1}{\sqrt{\mu}}\left(3U_5 - \chi U_4 - \sqrt{\mu}\mathrm{TOF}U_2\right) \tag{50}$$

B. Linearization of the Initial and Final States

Assume a trajectory with initial state $\boldsymbol{X}_1$ and final state $\boldsymbol{X}_2$. The differentials of the initial $\boldsymbol{X}_1$ and final states $\boldsymbol{X}_2$ are given as

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$$\mathrm{d}\boldsymbol{X}_1 = \delta\boldsymbol{X}_1 + \frac{\mathrm{d}\boldsymbol{X}_1}{\mathrm{d}t_1}\mathrm{d}t_1 \tag{51}$$

$$\mathrm{d}\boldsymbol{X}_2 = \delta\boldsymbol{X}_2 + \frac{\mathrm{d}\boldsymbol{X}_2}{\mathrm{d}t_2}\mathrm{d}t_2 \tag{52}$$

Expressing $\delta\boldsymbol{X}_2 = \Phi(t_2, t_1)\delta\boldsymbol{X}_1$ and solving Eq. (51) for $\delta\boldsymbol{X}_1$ yields

$$\mathrm{d}\boldsymbol{X}_2 = \Phi(t_2, t_1)(\mathrm{d}\boldsymbol{X}_1 - \dot{\boldsymbol{X}}_1\mathrm{d}t_1) + \dot{\boldsymbol{X}}_2\mathrm{d}t_2 \tag{53}$$

where $\Phi(t_2, t_1)$ is the state transition matrix (STM) forward in time (known as the navigation matrix [Battin [4] p. 461]) given previously in Eq. (45), whereas $\dot{\boldsymbol{X}}_1$ and $\dot{\boldsymbol{X}}_2$ are the time derivatives of the initial and final states, respectively. Expressing Eq. (53) in component form,

$$\mathrm{d}\boldsymbol{r}_2 = \boldsymbol{A}\mathrm{d}\boldsymbol{r}_1 + \boldsymbol{B}\mathrm{d}\boldsymbol{v}_1 - \boldsymbol{A}\dot{\boldsymbol{r}}_1\mathrm{d}t_1 - \boldsymbol{B}\dot{\boldsymbol{v}}_0\mathrm{d}t_1 + \dot{\boldsymbol{r}}_2\mathrm{d}t_2 \tag{54}$$

$$\mathrm{d}\boldsymbol{v}_2 = \boldsymbol{C}\mathrm{d}\boldsymbol{r}_1 + \boldsymbol{D}\mathrm{d}\boldsymbol{v}_1 - \boldsymbol{C}\dot{\boldsymbol{r}}_1\mathrm{d}t_1 - \boldsymbol{D}\dot{\boldsymbol{v}}_1\mathrm{d}t_1 + \dot{\boldsymbol{v}}_2\mathrm{d}t_2 \tag{55}$$

Similarly, for a trajectory propagated backward in time, the corresponding adjoint STM (known as the guidance matrix [Battin [4] p. 461]) is given by

$$\Phi(t_1, t_2) = \begin{bmatrix} \boldsymbol{P} & \boldsymbol{Q} \\ \boldsymbol{R} & \boldsymbol{S} \end{bmatrix} \tag{56}$$

and the component form of the differential $\mathrm{d}\boldsymbol{X}_2$ is given by

$$\mathrm{d}\boldsymbol{r}_1 = \boldsymbol{P}\mathrm{d}\boldsymbol{r}_2 + \boldsymbol{Q}\mathrm{d}\boldsymbol{v}_2 - \boldsymbol{P}\dot{\boldsymbol{r}}_2\mathrm{d}t_2 - \boldsymbol{Q}\dot{\boldsymbol{v}}_2\mathrm{d}t_2 + \dot{\boldsymbol{r}}_1\mathrm{d}t_1 \tag{57}$$

$$\mathrm{d}\boldsymbol{v}_1 = \boldsymbol{R}\mathrm{d}\boldsymbol{r}_2 + \boldsymbol{S}\mathrm{d}\boldsymbol{v}_2 - \boldsymbol{R}\dot{\boldsymbol{r}}_2\mathrm{d}t_2 - \boldsymbol{S}\dot{\boldsymbol{v}}_2\mathrm{d}t_2 + \dot{\boldsymbol{v}}_1\mathrm{d}t_1 \tag{58}$$

Given the symplectic nature of the Kepler STM, the forward STM $\Phi(t_2, t_1)$ and its adjoint $\Phi(t_1, t_2)$ are related as follows:

$$\Phi(t_1, t_2) = \Phi^{-1}(t_2, t_1) = \begin{bmatrix} \boldsymbol{P} & \boldsymbol{Q} \\ \boldsymbol{R} & \boldsymbol{S} \end{bmatrix} = \begin{bmatrix} \boldsymbol{D}^T & -\boldsymbol{B}^T \\ -\boldsymbol{C}^T & \boldsymbol{A}^T \end{bmatrix}$$

The time-of-flight differential is given as

$$\mathrm{dTOF} = \mathrm{d}t_2 - \mathrm{d}t_1 \tag{59}$$

which relates time of flight with the initial and final time partials.

C. Position and Time-of-Flight Partials

Each Lambert position or time-of-flight partial applies the appropriate constraints to Eqs. (54) and (55) for the forward in time differential, or Eqs. (57) and (58) for the backward in time differential. Substituting Eq. (54) into Eq. (55) and, similarly, Eq. (57) into Eq. (58) for the forward or backward in time differential, respectively, allows the derivation of the required partials. The time-of-flight differential relates the time of flight and the initial and final times with $\mathrm{dTOF} = -\mathrm{d}t_1$ and $\mathrm{dTOF} = \mathrm{d}t_2$ when $\mathrm{d}t_2 = 0$ and $\mathrm{d}t_1 = 0$, respectively. Tables 1 and 2 document the required Lambert solution partials, corresponding constraints, and the equivalent equation for the first method. Note that $\mathrm{d}\boldsymbol{X} = \delta\boldsymbol{X}$ when $\mathrm{d}t = 0$. For example, to determine the partial $[\partial\boldsymbol{v}_2/\partial\boldsymbol{r}_1]_{\mathrm{LT}}$, we apply the Lambert targeting problem constraints $\mathrm{d}\boldsymbol{r}_2 = 0$ and $\mathrm{d}t_1 = \mathrm{d}t_2 = 0$ to Eq. (54) and solve for $\mathrm{d}\boldsymbol{v}_1$:

$$0 = \boldsymbol{A}\mathrm{d}\boldsymbol{r}_1 + \boldsymbol{B}\mathrm{d}\boldsymbol{v}_1 \quad \mathrm{d}\boldsymbol{v}_1 = -\boldsymbol{B}^{-1}\boldsymbol{A}\mathrm{d}\boldsymbol{r}_1 \tag{60}$$

Applying the Lambert targeting constraints to Eq. (55), followed by a substitution in Eq. (60),

$$\mathrm{d}\boldsymbol{v}_2 = \boldsymbol{C}\mathrm{d}\boldsymbol{r}_1 + \boldsymbol{D}\mathrm{d}\boldsymbol{v}_1 = \boldsymbol{C}\mathrm{d}\boldsymbol{r}_1 + \boldsymbol{D}(-\boldsymbol{B}^{-1}\boldsymbol{A}\mathrm{d}\boldsymbol{r}_1) = (\boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A})\mathrm{d}\boldsymbol{r}_1$$

Hence, the required Lambert solution partial is obtained:

$$\left[\frac{\partial\boldsymbol{v}_2}{\partial\boldsymbol{r}_1}\right]_{\mathrm{LT}} = \boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A}$$

The other Lambert solution partials are obtained in a similar manner and are tabulated in Tables 1 and 2. Note that various symmetries identified in the previous section are easily verified from these expressions. The last columns, labeled "Partial IDs" give a unique

Table 1 Position partials from Kepler STM method

Lambert solution partial	Constraints	Equivalent equation	Partial ID[a,b]
$[\partial\boldsymbol{v}_2/\partial\boldsymbol{r}_1]_{\mathrm{LT}} = \boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A}$	$\mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_1 = \mathrm{d}t_2 = 0$	Eq. (34) ($\boldsymbol{p} = \boldsymbol{r}_1$)	31–39
$[\partial\boldsymbol{v}_2/\partial\boldsymbol{r}_2]_{\mathrm{LT}} = \boldsymbol{D}\boldsymbol{B}^{-1}$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}t_1 = \mathrm{d}t_2 = 0$	Eq. (34) ($\boldsymbol{p} = \boldsymbol{r}_2$)	40–48
$[\partial\boldsymbol{v}_1/\partial\boldsymbol{r}_2]_{\mathrm{LT}} = \boldsymbol{R} - \boldsymbol{S}\boldsymbol{Q}^{-1}\boldsymbol{P} = -(\boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A})^T$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}t_1 = \mathrm{d}t_2 = 0$	Eq. (33) ($\boldsymbol{p} = \boldsymbol{r}_2$)	22–30
$[\partial\boldsymbol{v}_1/\partial\boldsymbol{r}_1]_{\mathrm{LT}} = \boldsymbol{S}\boldsymbol{Q}^{-1} = -(\boldsymbol{B}^{-1}\boldsymbol{A})^T$	$\mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_1 = \mathrm{d}t_2 = 0$	Eq. (33) ($\boldsymbol{p} = \boldsymbol{r}_1$)	13–21

[a]See Table 3.
[b]ID denotes identification.

Table 2 TOF partials from Kepler STM method

Lambert solution partial	Constraints	Equivalent equation no.	Partial ID[a,b]
$[\partial\boldsymbol{v}_2/\partial\mathrm{TOF}]_{\mathrm{LT}} = -[\partial\boldsymbol{v}_2/\partial t_1]_{\mathrm{LT}} = (\boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A})\dot{\boldsymbol{r}}_1$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_2 = 0$	Eq. (36)	7–9
$[\partial\boldsymbol{v}_2/\partial\mathrm{TOF}]_{\mathrm{LT}} = [\partial\boldsymbol{v}_2/\partial t_2]_{\mathrm{LT}} = \dot{\boldsymbol{v}}_2 - \boldsymbol{D}\boldsymbol{B}^{-1}\dot{\boldsymbol{r}}_2$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_1 = 0$	Eq. (36)	10–12
$[\partial\boldsymbol{v}_1/\partial\mathrm{TOF}]_{\mathrm{LT}} = [\partial\boldsymbol{v}_1/\partial t_2]_{\mathrm{LT}} = -(\boldsymbol{R} - \boldsymbol{S}\boldsymbol{Q}^{-1}\boldsymbol{P})\dot{\boldsymbol{r}}_2 = (\boldsymbol{C} - \boldsymbol{D}\boldsymbol{B}^{-1}\boldsymbol{A})^T\dot{\boldsymbol{r}}_2$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_1 = 0$	Eq. (35)	4–6
$[\partial\boldsymbol{v}_1/\partial\mathrm{TOF}]_{\mathrm{LT}} = -[\partial\boldsymbol{v}_1/\partial t_1]_{\mathrm{LT}} = -(\dot{\boldsymbol{v}}_1 - \boldsymbol{S}\boldsymbol{Q}^{-1}\dot{\boldsymbol{r}}_1) = -(\dot{\boldsymbol{v}}_1 + (\boldsymbol{B}^{-1}\boldsymbol{A})^T\dot{\boldsymbol{r}}_1)$	$\mathrm{d}\boldsymbol{r}_1 = \mathrm{d}\boldsymbol{r}_2 = \mathrm{d}t_2 = 0$	Eq. (35)	1–3

[a]See Table 3.
[b]ID denotes identification.

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Table 3 Reference case partial derivative values

Partial ID[a]	Direct approach D	STM approach S	Central difference F	$\frac{D-S}{D}$	$\frac{D-F}{D}$
1	0.14275266177719367	0.14275266177719104	0.14275266179498505	1.84709663E − 14	−1.24630806E − 10
2	−0.25683817511481571	−0.25683817511481111	−0.25683817512680707	1.79390211E − 14	−4.66883580E − 11
3	5.2570523571927354E − 004	5.2570523571926595E − 004	5.2570523577663466E − 004	1.44366364E − 14	−1.09112712E − 10
4	−1.2131567709212115E − 002	−1.2131567709212396E − 002	−1.2131567712673092E − 002	−2.31647892E − 14	−2.85286822E − 10
5	−0.28646599208828566	−0.28646599208827839	−0.28646599209268608	2.53850760E − 14	−1.53610718E − 11
6	1.5414843376598841E − 003	1.5414843376598437E − 003	1.5414843377408866E − 003	2.61646004E − 14	−5.25483650E − 11
7	−0.14275266177719367	−0.14275266177719212	−0.14275266179498505	1.08881486E − 14	−1.24630806E − 10
8	0.25683817511481571	0.25683817511481188	0.25683817512680707	1.49131619E − 14	−4.66883580E − 11
9	−5.2570523571927354E − 004	−5.2570523571926139E − 004	−5.2570523577663466E − 004	2.30986173E − 14	−1.09112712E − 10
10	1.2131567709212115E − 002	1.2131567709211755E − 002	1.2131567712673092E − 002	2.97424439E − 14	−2.85286822E − 10
11	0.28646599208828566	0.28646599208828016	0.28646599209268608	1.91841409E − 14	−1.53610718E − 11
12	−1.5414843376598841E − 003	−1.5414843376598524E − 003	−1.5414843377408866E − 003	2.05378046E − 14	−5.25483650E − 11
13	−0.87581965775919213	−0.87581965775919079	−0.87581965772352821	1.52116667E − 15	4.07206178E − 11
14	0.59373435674742359	0.59373435674742336	0.59373435671293884	3.73979710E − 16	5.80811059E − 11
15	5.9995779046728035E − 003	5.9995779046727975E − 003	5.9995779139343404E − 003	1.01199325E − 15	−1.54369806E − 09
16	0.59373435674742359	0.59373435674742236	0.59373435676844999	2.05688843E − 15	−3.54138212E − 11
17	2.2675551310063654E − 002	2.2675551310064507E − 002	2.2675551348785010E − 002	−3.76389492E − 14	−1.70762582E − 09
18	4.2494900210715341E − 004	4.2494900210716859E − 004	4.2494899229398480E − 004	−3.57191829E − 14	2.30925803E − 11
19	5.9995779046728035E − 003	5.9995779046727940E − 003	5.9995779055642996E − 003	1.59027503E − 15	−1.48593138E − 10
20	4.2494900210715341E − 004	4.2494900210717640E − 004	4.2494900226864500E − 004	−5.40890473E − 14	−3.80025816E − 10
21	0.74849340207512749	0.74849340207512771	0.74849340195161651	−2.96655396E − 16	1.65012781E − 10
22	−0.11464346247357682	−0.11464346247357948	−0.11464346245505376	−2.32419302E − 14	1.61570993E − 10
23	0.24709546240540511	0.24709546240540892	0.24709546241611888	−1.53888455E − 14	−4.33588027E − 11
24	−7.7909114808782567E − 003	−7.7909114808782619E − 003	−7.7909115017984181E − 003	−6.67979658E − 16	−2.68520073E − 09
25	−5.1127505015577390E − 002	−5.1127505015577057E − 002	−5.1127505024032092E − 002	6.51443680E − 15	−1.65365041E − 10
26	−0.74246530945269806	−0.74246530945269296	−0.74246530946336020	6.87847081E − 15	−1.43604529E − 11
27	−2.5025470706876427E − 003	−2.5025470706876492E − 003	−2.5025470842190475E − 003	−2.59943675E − 15	−5.40705303E − 09
28	−6.2565606746204615E − 003	−6.2565606746204511E − 003	−6.2565606755541096E − 003	1.66358827E − 15	−1.49227061E − 10
29	−4.1648802470078666E − 003	−4.1648802470079307E − 003	−4.1648802468912238E − 003	−1.54109513E − 14	2.80062813E − 11
30	−1.2842068520343359	−1.2842068520343362	−1.2842068519668488	−1.72904081E − 16	5.25516020E − 11
31	0.11464346247357682	0.11464346247357948	0.11464346247170711	−2.32419302E − 14	1.63089351E − 11
32	5.1127505015577390E − 002	5.1127505015577057E − 002	5.1127505033399599E − 002	6.51443680E − 15	−3.48583579E − 10
33	6.2565606746204615E − 003	6.2565606746204511E − 003	6.2565606777225139E − 003	1.66358827E − 15	−4.95807950E − 10
34	−0.24709546240540511	−0.24709546240540892	−0.24709546241056776	−1.53888455E − 14	−2.08933339E − 11
35	0.74246530945269806	0.74246530945269296	0.74246530946856437	6.87847081E − 15	−2.13697636E − 11
36	4.1648802470078666E − 003	4.1648802470079307E − 003	4.1648802321027053E − 003	−1.54109513E − 14	3.57877306E − 09
37	7.7909114808782567E − 003	7.7909114808782619E − 003	7.7909114821093058E − 003	−6.67979658E − 16	−1.58010924E − 10
38	2.5025470706876427E − 003	2.5025470706876492E − 003	2.5025470705580997E − 003	−2.59943675E − 15	5.17644920E − 11
39	1.2842068520343359	1.2842068520343362	1.2842068519158045	−1.72904081E − 16	9.22993071E − 11
40	0.44501264306543264	0.44501264306543220	0.44501264295322618	9.97924879E − 16	2.52142196E − 10
41	1.1628008100243965	1.1628008100243965	1.1628008101155329	0.00000000	−7.83766524E − 11
42	−1.6679263381085267E − 002	−1.6679263381085271E − 002	−1.6679263369924882E − 002	−2.08009600E − 16	6.69117373E − 10
43	1.1628008100243965	1.1628008100243965	1.1628008099347746	0.00000000	7.70741318E − 11
44	0.46967289367853671	0.46967289367853426	0.46967289359217246	5.20040791E − 15	1.83881702E − 10
45	−1.6481622762208353E − 002	−1.6481622762208319E − 002	−1.6481622777702574E − 002	2.10503972E − 15	−9.40090672E − 10
46	−1.6679263381085267E − 002	−1.6679263381085267E − 002	−1.6679263381764370E − 002	−0.00000000	−4.07153825E − 11
47	−1.6481622762208353E − 002	−1.6481622762208326E − 002	−1.6481622764518675E − 002	1.68403173E − 15	−1.40175649E − 10
48	−1.4737961685886782	−1.4737961685886787	−1.4737961682197132	−3.01323368E − 16	2.50350130E − 10

[a]See Tables 1 and 2 for partial identification (ID) definition.

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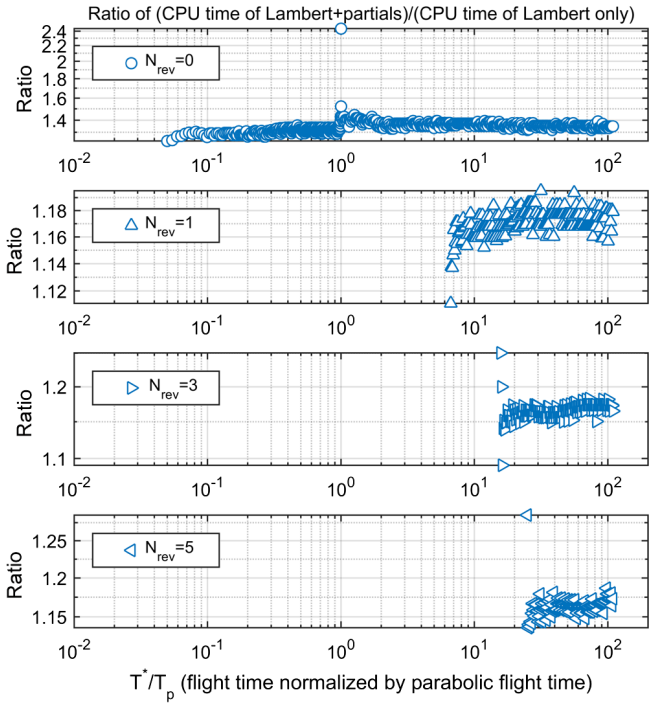


Fig. 3 Direct approach results: computing costs for Lambert plus partials.

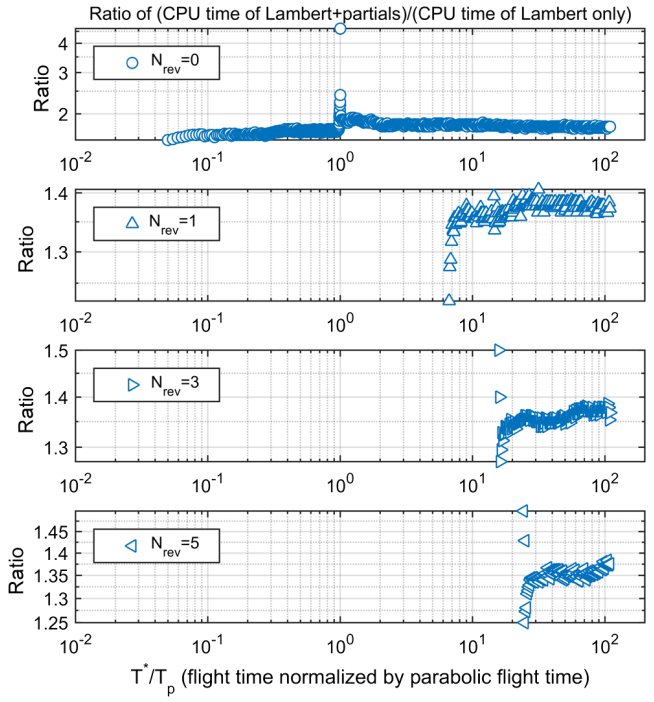


Fig. 4 STM approach results: computing costs for Lambert plus partials.

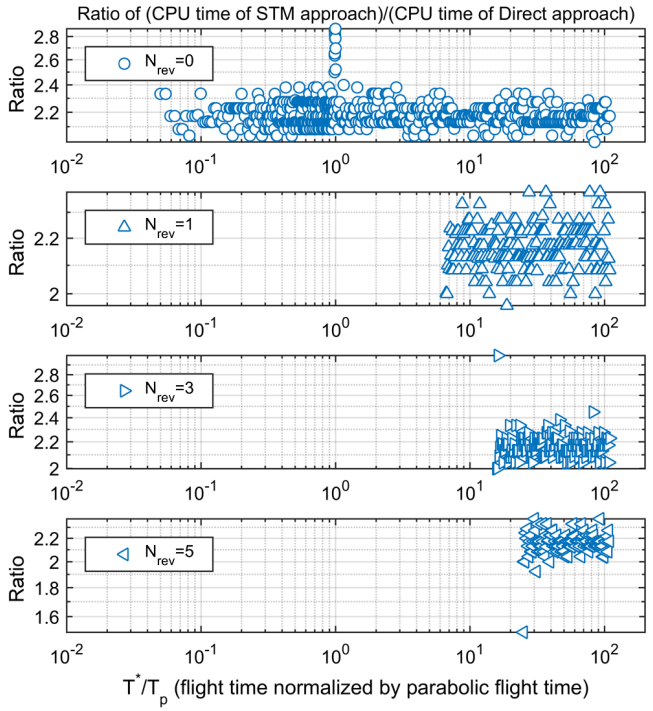


Fig. 5 Relative computing times comparing the STM and direct approaches.

identifier for each partial derivative and are useful for correlating the evaluated expressions in the numerical example presented in Table 3. Note, that $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{D}$ are not independent ([4] p. 459) per the identity $\boldsymbol{D} * \boldsymbol{A}^T - \boldsymbol{C} * \boldsymbol{B}^T = \boldsymbol{I}$. Therefore, one can avoid evaluating the expensive expression for $\boldsymbol{C}$ in Eq. (48), and many of the expressions in Tables 1 and 2 can be further simplified, leading to improved efficiency. The timings in the following section do not take advantage of this identity. Future works will consider such simplifications, among other improvements such as second order terms and the application to nonsymplectic systems.

V. Simulations and Performance

In this section, the performance of the two methods is evaluated on a test set of Lambert problems using randomized inputs. The test is devised in order to isolate the performance of each method according to bins of solution type (i.e., hyperbola, parabola, zero-revolution ellipse, and multirevolution ellipse). For estimating the cost of computing the partials in Eqs. (33) and (34) (direct approach), a computationally efficient form is implemented in Fortran alongside the original code from Ref. [1]. The code is compiled using the Intel Fortran compiler, version 15.0, subject to the "-fast -xHost" optimization flag. The test hardware used for the simulations comprised an Intel Xeon E5-2630 CPU at 2.60 GHz, running on an X86-64 Linux workstation.

Figure 3 shows the ratio of runtimes for the Lambert-plus-partial computation vs the Lambert-only case, as a function of T^*/T_p, where T_p is the parabolic time for each input geometry. Each data point represents 200,000 Lambert cases, where each component of the input vectors ($\boldsymbol{r}_1$ and $\boldsymbol{r}_2$) varies randomly between -10 and 10 in a uniform distribution when expressed in normalized units ($r_1 = 1$ and

Table 4 Computing time summary (average cost beyond the Lambert solution cost)

Parameter	Hyperbola, %	Parabola, %	Zero-revolution ellipse, %	Multirevolution ellipse, %
Direct approach	30	140	35	15
STM approach	65	450	80	35

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Table 5 Reference case inputs (normalized units)

Parameter	Value
$\boldsymbol{r}_1$	[−0.66234662571997105, 0.74919751798749190, −1.6259997018919074E − 004]
$\boldsymbol{r}_2$	[−0.24986234273434585, −0.69332384278075210, 4.959901216866255 1E − 003]
T^*	2.5803148345055149
N_{rev}	0
μ	1
Transfer type	Direct

$\mu = 1$). The ratio TOF/T_p is varied from 0.05 to ∼100, thereby allowing hyperbolic ($T^*/T_p < 1$), parabolic ($T^*/T_p = 1$), and elliptical zero-revolution and multirevolution ($T^*/T_p > 1$) solutions. The target flight time T^* varies between 0.01 and 3000 TU when expressed in normalized units ($r_1 = 1$ and $\mu = 1$). The variable N_{rev} denotes a fixed number of Lambert revolutions. The cost of computing the partials is fixed, whereas the cost of computing the Lambert solution varies as a function of T^*/T_p. This dependency is particularly evident in Fig. 3 near the parabolic case ($T^*/T_p = 1$). For the zero-revolution and hyperbolic cases, the extra computing efforts for obtaining the partials are approximately 35 and 30%, respectively. For the multirevolution case, this number varies between 10 and 20%, depending upon the ratio T^*/T_p and N_{rev} (see Fig. 3). The Lambert routine itself varies in runtime because the number of iterations necessary to converge varies from case to case (usually between two and four iterations). Figure 4 shows a similar plot for the timings of the second method based on the Kepler STM. The STM approach for computing the partials is slower than the direct approach by a factor of ∼2.2 (see Fig. 5) on average. Table 4 summarizes the simulation results presented in Figs. 3–5.

The extra computational cost of computing $\partial \boldsymbol{y}/\partial \boldsymbol{p}$ via a central difference method is 1400%, noting that 14 additional Lambert calls are necessary. If the application requires instead $\partial \boldsymbol{y}/\partial \boldsymbol{w}$, where $\boldsymbol{p} = \boldsymbol{p}(\boldsymbol{w})$ and $\boldsymbol{w}$ is some parameter vector of length n that specifies $\boldsymbol{p}$, then $\partial \boldsymbol{y}/\partial \boldsymbol{w}$ can be computed via the chain rule as $(\partial \boldsymbol{y}/\partial \boldsymbol{p})(\partial \boldsymbol{p}/\partial \boldsymbol{w})$ using the direct or STM approaches outlined in this study. On the other hand, a central difference computation of $\partial \boldsymbol{y}/\partial \boldsymbol{w}$ can be directly achieved using only $2n$ additional Lambert calls. An example $\boldsymbol{w}$ constituent is an ephemeris epoch that specifies the location of a celestial body. If the full $\boldsymbol{p}$ vector is free in an optimization context, such as the case for the selection of a maneuver location, then the full $\partial \boldsymbol{y}/\partial \boldsymbol{p}$ must be computed. In addition to the extra computing time, the finite difference methods are limited in accuracy due to precision and roundoff errors, and they require tuning for step sizes, whereas both of the presented approaches achieve accuracy near to that of the computed velocity vectors. Hence, the analytic partials presented in this study provide up to an order of magnitude in speed benefit while achieving high accuracy.

For reference, Table 3 provides the resulting values of the partials from the direct, STM, and central difference approaches for the set of inputs specified in Table 5. The step size used for the central difference approximations is 10^{-5}.

VI. Conclusions

Two methods to derive partial derivatives of the outputs with respect to the inputs of the Lambert boundary value problem are presented. The first approach is based on a recently proposed vercosine Lambert formulation, whereas the second approach uses the Keplerian state transition matrix. The STM is computed in terms of Battin's fundamental perturbation matrices ([4] p. 467). For both approaches, the first-order partials of the two velocity vectors with respect to the two position vectors and the times are derived and documented. The partials from the first approach, expressed in a computationally efficient form, typically require ∼15 to 35% more computation time over the Lambert solution alone. The STM-based approach is approximately twice as slow (on average) when compared to the first approach, but it is general and can be applied to other Lambert algorithms or other non-Keplerian targeting problems. Compared to the central differencing or complex step approach, both newly presented methods are approximately an order of magnitude faster. The accuracy of the partials is limited only by the accuracy of the underlying Lambert solutions (both methods) and the computed STM (second method only).

The rapid access to partial derivatives of the Lambert problem solution allows for fast trajectory design techniques that implicitly enforce constraints via embedded Lambert problems, avoiding the need for shooting methods.

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